



ICT Capstone Project Expo

Wednesday, 28 May, 10.00am–12.00pm

Studio Theatre space (next to Stanley Burbury Theatre)

Staff, students, alumni, and industry members are invited to celebrate the work of our KIT300 Capstone Project teams. See live demos of student-made mobile apps, websites, and games!

[View the Bachelor of ICT
Course Handbook](#)

[Take the Attendee Survey](#)

UNIVERSITY of
TASMANIA 

CRICOS 00586B

KIT300 Capstone Projects

The capstone project unit provides an opportunity for students to consolidate the knowledge and skills they have acquired in their course and apply them to a substantial ICT project. Students extend their knowledge and skills in the areas of project management and system development through completing professional project documentation and managing the development of the team project using the agile methodology.

These projects have been developed over the course of **one semester** (formerly two semesters). Students work in teams with an industry client.

If you are interested in being a client in future semesters, please contact Quan.Bai@utas.edu.au (BICT) or Ming.Wong@utas.edu.au (Masters).

Unit Coordinator: Quan.Bai@utas.edu.au

[View KIT300 Unit Info](#)

[View KIT700 Unit Info](#)

Tasmanian Natural Hazards Web Atlas

[Poster](#)

Rahul Desai, Samuel Eleveld, Nimra Iqbal, Ibrar Majeed, Dominic Rose

A functional and user-friendly web platform allowing end users to view climate data on an interactive map as a graphical overlay, with different colours mapped to different values for the given environmental variable. Insights into the provided data, a guide for new users, and a glossary of climate terms are built into the platform.

Interactive QR for Smart Recycling

[Poster](#)

Tristan Harris, Athen Jumper, Hasibur Rahman, Xavier Scott, Kentaro Watts, Hongchao Xue

A mobile app for tracking recycling submissions, linked to bins via a QR code system. Includes functionality for moderators and administrators to track bin usage across all users.

Easy Price Tracker Mobile App

[Poster](#)

Angus Cooper, Mohamed Kariapper, Patrick Lennon, Cam Munro, Tyson Light, Kai Jun Wong

Price tracking on products from supported websites with functionality for paid and trial users. ReCAPTCHA integration and Google Passkey login. Email system to notify users of important updates.

<https://easypricetracker.com>

Agreements Microservice

[Poster](#)

Bronagh Falloon, Steph Hall, Cameron Nalder, Callum O'Rourke

Application that creates agreement templates and can send them out via email services.

Face in the Space

[Poster](#)

Sumaiya Aktar, Rafid Al Azad, Elliot Franklin, Matthew Laughlin, Zarif Sattar Khan

Face in the Space is a prototype music learning experience aimed at teaching younger audiences' musical notation and theory through gamification.

<https://faceinthespace0.itch.io/faceinthespacedemo>

Esoteroglyph

[Poster](#)

Will Bruce, Jake Rohrlach, Darius Rosa-Denham, Cade Swanson

As a well-travelled archaeologist, you've probably seen your fair share of puzzling sites, but you're sure you've never seen something like this before. Ruins come to life on a mysterious island in the Mediterranean Sea, and it's your job to catalogue the effects and abilities of the logic-defying glyph stones scattered across the island's surface.

<https://d00by.itch.io/esoteroglyph>

Computer Vision Roof Defect Detection

[Poster](#)

Jack Duffy, Vivek Jolly, Lai Phuc Anh Do, Josh Sideris

Application which uses Computer Vision to find and label detected defects in roofing images.

Booking Management for Event Bookings

[Poster](#)

Rifatul Bari, Iffat Jahan Borna, Nitish Gaire, Mahmood Kardar, Miraj Poudel

An intuitive and efficient platform that empowers merchants to manage their services, events, bookings, and client communication all in one place. Designed for ease of use and control, the module simplifies day-to-day operations and enhances the overall merchant experience.

IPA Name Translation and Speech Synthesis

Leesa Fernando, Gurleen Kaur, Midori Sugie, Caleb Wittmer

An easy to access and use Text-To-Speech software tool that helps users pronounce various names from different nationalities.

KIT109 Games Fundamentals Projects

Students learn the basics of game development technology and game design, via an intensive process learning practical skills for developing games in the Unity game engine. They then have around **6 weeks** to produce their own 2D game.

Unit Coordinator: Ian.Lewis@utas.edu.au

[View Unit Info](#)

[Play All Games on ITCH.io](#)

[Games @ UTAS YouTube Video](#)

[Games @ UTAS Part 2 YouTube Video](#)

UTAS “GOAT” Collection

Some of the best games from previous years of KIT109 Games Fundamentals.

Extrication

Ethan Duffin

Extrication is a hostage-rescue top-down shooter that pits you alone against a global crime syndicate.

<https://ugly2.itch.io/extrication>

Wings and Whiskers

Laura Glover

A cute local co-op puzzle-platformer starring an unlikely duo: Wings the Pigeon and Whiskers the Ferret. Embark on a journey across the lush landscapes of Daisy Banks as you face a range of obstacles that can only be conquered through their combined skills.

<https://cakeadooples.itch.io/wings-whiskers>

Fur Force – Kittens on Call

Matthew Monty

Using teamwork, agility... and trusty water cannons, players must put out fires, solve puzzles and navigate the trickiest of environments in this 2D action-platformer. Help the Fur Force team on their mission to save a poor mother and her baby kittens from the arsonist humans!

<https://woofleboofle.itch.io/fur-force-kittens-on-call>

Soup Weather

Brie Ratten

Soup Weather is a narrative exploration game in which you play as a borderline mentally deranged, but sort of normal human named Bean. Learn about Bean's life and relationships through strange conversations, weird inner dialogue, and nonsensical mini games as they slip further into their own delusions.

<https://beemjerky.itch.io/soup-weather>

Outpost Zero

Isaak Cole

You're a soldier sent on a mission gone wrong to investigate a secret moon research base, all the scientists are already dead when you arrive, and there's a mysterious blue substance everywhere. With your mission compromised, you'll need to fight, jump, and blast your way across the desolate base and moon to discover what happened, gathering evidence for your employer, and attempting to reach your ship and escape.. if you can.

<https://ibcole.itch.io/outpost-zero>



KIT207 Games Design and Production Projects

Students work in teams to design and develop a prototype of a 3D game in the Unity game engine. Students learn skills that are essential for the game designer, such as: 3D modelling, rapid prototyping tools, level editing tools, and scripting.

Unit Coordinator: Ian.Lewis@utas.edu.au

[View Unit Info](#)

Fantasy Lab Escape by Otherworldly Analysts

Harry Beumer, John Streat, Josh Daniels, Ryan Boulter

You're an experiment. After years of being locked in stasis, you awaken in an abandoned lab. Escape by fighting through hordes of experiments and upgrading your way to the surface.

<https://josh-daniels.itch.io/fantasy-lab-escape>

War Watchers by Dungeoneers

James Coady, Kler Mu, Patrick Wing, Liam Patridge

In Warwatchers, you play as a fledgling brigadier gunning for a spot on the legendary band of Warwatchers! When a rebellion breaks out amongst nearby planets, its your chance to prove yourself worthy of the spot!

<https://j-coady.itch.io/kit-207-game>

KIT305/721 Mobile Application Development Projects

Students create a Native Android App (Kotlin), Native iOS App (Swift), and Cross-Platform App (Flutter) on an assignment theme over the course of 13 weeks.

The theme this semester was a live sports stat-tracking application, which is capable of recording actions and scores as they happen during a live match. Students are challenged to employ design principles and consider usability goals in the creation of their apps.

Unit Coordinator: Lindsay.Wells@utas.edu.au

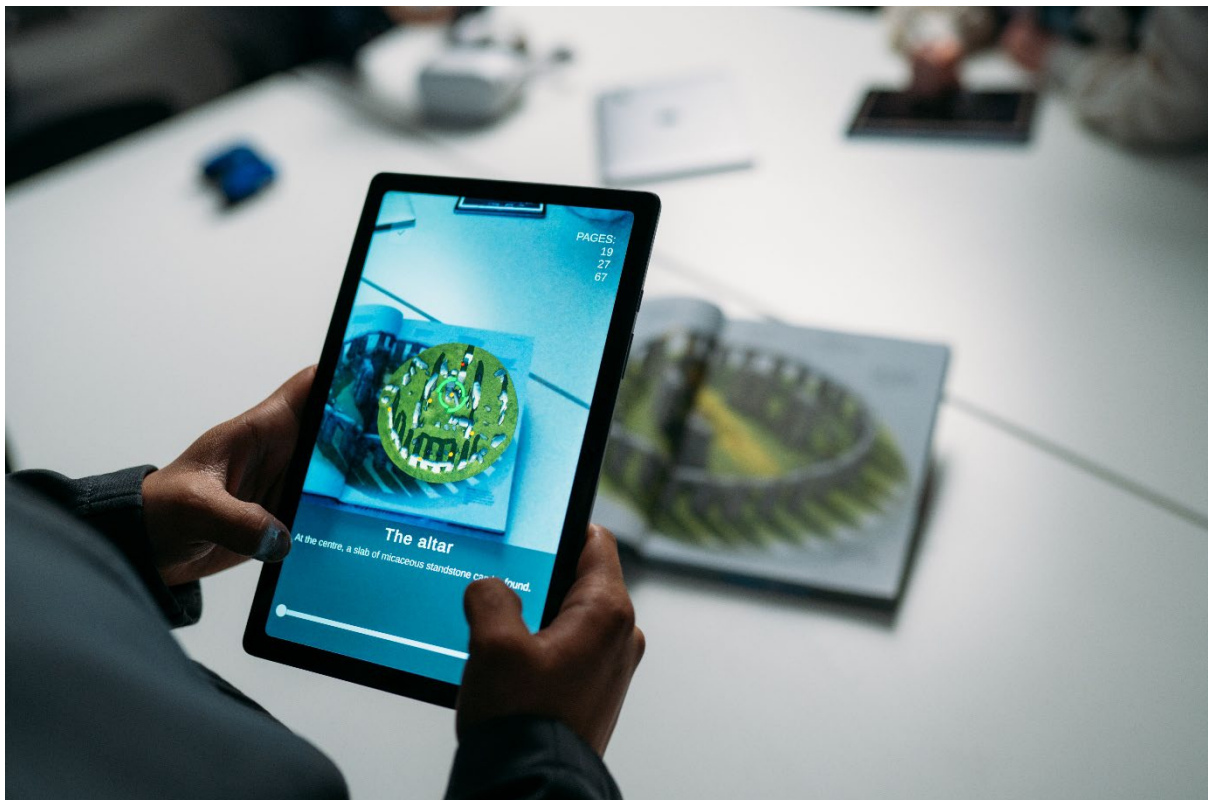
[View Unit Info](#)

Onnida Hempattawee (Volleyball; Android and iOS)

Yu Wang (AFL; Android)

Xi Chen (AFL; Android)

Giang Nguyen (Soccer; Android and iOS)





KIT208 Virtual and Mixed Reality Projects

Students work in teams over the course of 6 weeks to create an Augmented Reality (AR) or Virtual Reality (VR) application prototype that works on real hardware (Android Phone or Meta Quest VR Headset). Students produce an application that is designed to tackle a real-world problem.

Unit Coordinator: Winyu.Chinthammit@utas.edu.au

[View Unit Info](#)

[Watch on YouTube](#)

Antarctic Interactive Map

[Report](#)

Ross Vandenberg, Kristan Gorry, James Kidd

This is an application designed to get students and young people engaged in the Antarctic experience. It uses AR technology to show the degradation of the ice sheets in the Antarctic on a sliding scale throughout the years as well as show points of interest (such as animal habitats) through pins on the 2D map.

Carbon Emission Education

[Report](#)

Emiliano Balseira Martinez, Christopher Snodgrass, Caleb Wittmer

This application aims to provide an experience that encapsulates the abstract concepts of carbon emissions into an easily understood format. This is achieved by providing the user with various points of interaction designed to inform how changes to aspects of a household affect their carbon footprint.

Bottle Battlers

[Report](#)

Oliver Connor, Fabian Lampasona, Micah Sanderson

Have you or anyone you know ever struggled to put the cask down? It is easy to let the train gain speed once it leaves the station, but sometimes you have to hit the brakes! This is exactly what this VR application intends to do. Bottle Battlers: Deliverance targets such behaviours by drilling healthy alternatives. Bottle Battlers - "Get your liver delivered!"

Climate Change Education

[Report](#)

Jordan Holloway, Matthew Dunn

This application aims to provide an experience that encapsulates the abstract concepts of carbon emissions into an easily understood format. This will be achieved by providing the user with various points of interaction designed to inform how changes to aspects of a household affect their carbon footprint.

Tank Operations & Advanced Safety Training

[Report](#)

Noah West, Josh Rayner

The Tank Operations and Advanced Safety Training v2 (TOAST2) application allows companies operating tank farms to train new pipeline operators with ease. It reduces the costs associated with ordinary training methods and improves accessibility of training, while being undertaken in a hazard-free environment. This access to training will reduce the risk to global infrastructure and the developed/developing nations that rely upon it.

AR Wood Heater Previewer

[Report](#)

Jake Rohrlach, Ben Dolliver, Shawn Hatten

Working in wood heater installation often presents the challenge of visualising what a wood heater will look like in a location without the physical heater in place. Using an AR application to visualise the heater solves this problem and could save up to four hours on each installation.

Overcoming Avoidance Behaviours

[Report](#)

Patrick Wing, Nick Dexter, William Ditchfield

The application serves to help users confront their fears and overcome exhibited avoidance behaviours through a series of scenarios. These scenarios gradually increase in intensity to help the user slowly expose themselves to the situations in a controlled environment.

Crane Training (2022)

Zachary Berryman

The new era of VR Crane Training. With no risk, consequences or experience. Learn with an immersive experience that replicates a real Tower Crane operator in a virtual construction environment.

<https://xepries.itch.io/kit-208-724-assignment-4-5>

What Does Christmas Look Like (Colour-blindness Simulator; 2022)

Jasper Lennaen, Toby Coy, Johannes Nicholas, Delaney Roach, Brayden Ransom-Frost

Colour Vision Deficiency (CVD) affects 1 in 12 males and 1 in 200 females (Collinge, 2017). 'What Does Christmas Look Like' (WDCLL) is a virtual reality (VR) application that allows colour-able designers to experience the effects of various kinds of CVD whilst editing and recolouring the virtual environment to best suit a wide range of colour-able and colour deficient people.

<https://seasideview.itch.io/kit208-colour-blindness>

High Voltage Powerline Training Simulator (2022)

Harry Driessen, Oliver Lohrey, Imran Aryan Kamil

High-voltage powerline work is very dangerous, especially so for inexperienced trainees. This application provides a safe and affordable way for high-voltage powerline workers to learn and practice real-world situations such as installing a spacer on a set of power lines.

<https://hdriessen.itch.io/high-voltage-power-line-vr-simulator>

KIT307 Computer Graphics and Animation Videos

Students develop a short movie using procedural animation and texture and lighting effects. Knowledge and skills needed include 3D coordinate systems, 3D transformations, scene graphs, key-frame animation, lighting and shading.

The entire animation is programmed using computer graphics techniques applied using the Unity game engine. Since the animation is largely generated through code and not traditional 3D graphics software like Blender or 3DS Max, **the outcomes showcase a more technical feat, rather than a purely aesthetic one.**

Unit Coordinator: Robert.Ollington@utas.edu.au

[View Unit Info](#)

[Watch on YouTube](#)

Jes Lewincamp

Lily Gough

Nathan Johns

Eliza Gurney

Reuben Clark

Charlie Cooper

Ethan Fletcher

Jack Perry

We acknowledge the Palawa/Pakana and Gadigal people, the Traditional Custodians of the land upon which we live and work. We honour their enduring culture and knowledges as vital to the self-determination, wellbeing and resilience of their communities.